AETHER MK-1: AMBIENT EDGE-COMPUTING TOPOLOGY FOR HUMAN EMERGENCY RESCUE

A PROJECT REPORT

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INTERNAL EXAMINER

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ABSTRACT

During structural collapses such as earthquakes and industrial accidents, the critical window to locate trapped survivors is 72 hours. Traditional Search and Rescue (SAR) technologies have severe operational limitations: thermal cameras cannot penetrate solid concrete, rescue dogs are impeded by environmental hazards, and acoustic microphones require the victim to be conscious and vocal.

Project AETHER MK-1 is a non-line-of-sight human detection system using standard 2.4GHz Radio Frequency (RF) waves. By deploying a localized Wi-Fi mesh network over a debris field, AETHER utilizes a TinyML neural network to analyze the multipath interference caused by the physical movements of trapped bodies, allowing rescue teams to identify survivor locations through solid barriers.

The MK-1 prototype operates as a binary classifier, utilizing Received Signal Strength Indicator (RSSI) telemetry to differentiate between static environments (Dead Rubble) and dynamic human presence (Trapped Survivor). The complete system was built for under ₹1,000, making it mass-deployable by the NDRF within the critical 72-hour survival window with zero cloud dependency.

1. INTRODUCTION

1.1 Project Overview

Structural collapses create chaotic debris environments where every minute is critical. Current search-and-rescue technology depends on either victim cooperation (acoustic sensors) or line-of-sight access (cameras, thermal imaging), both of which fail in the most critical scenarios when a survivor is trapped, unconscious, or buried under reinforced debris.

AETHER MK-1 addresses this gap by leveraging the ubiquitous 2.4GHz Wi-Fi band as an active radar system. RF waves at this frequency penetrate through wood, drywall, and non-reinforced concrete. A human body inside the radiation field causes measurable disruptions to the signal's multipath propagation. By training a TinyML neural network on these disruption patterns, AETHER can classify whether a location contains living human movement or is purely static debris, without requiring the victim to cooperate in any way.

The platform is built on the DOIT ESP32 DevKit V1, a commodity Wi-Fi microcontroller available for under ₹500. This enables mass deployment: a rescue team can flood an entire disaster zone with sensor nodes for less than the cost of a single conventional rescue robot. The AI inference runs entirely on the edge, requiring no internet, no servers, and no cellular infrastructure.

2. SYSTEM ANALYSIS

2.1 Existing System — Limitations

The existing search and rescue systems have severe operational limitations that prevent effective deployment in real disaster scenarios:

Technology	Limitation in Disaster Scenarios
Thermal Cameras	Cannot penetrate solid concrete or thick debris layers. Thermal radiation is blocked by dense materials, making it impossible to detect body heat through rubble.
Rescue Dogs	Blocked by fire, toxic smoke, and chemical hazards. Availability is limited and large-scale deployment requires significant logistics time.
Acoustic Microphones	Require the victim to be conscious and actively producing sound. Completely ineffective for unconscious survivors who constitute the majority of time-critical cases.
PIR Sensors	Detect thermal radiation emitted by the human body. Cannot penetrate solid concrete or thick debris. Completely useless in structural collapse scenarios.
Camera Robots	Fail in darkness, dense smoke, and non-line-of-sight environments. Debris fields are inherently obstructed, making optical approaches unreliable.

KEY FINDING: Every existing system requires either line-of-sight OR victim cooperation. AETHER MK-1 requires neither.

2.2 Proposed System — AETHER MK-1 Advantages

Advantage	Description
RF Wave Penetration	2.4GHz signals penetrate wood, drywall, and non-reinforced barriers. Optical visibility is completely irrelevant.
Passive Detection	Detects movement passively. Victim does not need to be conscious, vocal, or cooperative in any way.
Full Edge Computing	TinyML inference runs entirely on the ESP32 microcontroller. Zero cloud infrastructure required.
Ultra-Low Cost	Complete node built for under ₹1,000. Hundreds deployable across an entire disaster zone.
Zero Infrastructure	Operates independently of cellular towers, internet, and power grids.
Instant Binary Answer	Dead Rubble vs Trapped Survivor. Immediate proof of life for rescue coordinators.

3. HARDWARE & SOFTWARE CONFIGURATION

3.1 Hardware Components & Budget Profile

A primary objective of AETHER MK-1 is achieving ultra-low deployment costs. The complete edge-computing radar system was built for under ₹1,000.

Component	Specification	Role in AETHER
DOIT ESP32 DevKit V1 (x2)	2.4GHz Wi-Fi + BT SoC, 240MHz dual-core	TX Illuminator + RX AI Brain (under ₹500 each)
5mm Red LED	Standard through-hole LED, forward voltage 2V	Physical alarm indicator
330Ω Resistor	Carbon film, 1/4W, tolerance ±5%	Current limiting for LED circuit
Solderless Breadboard	830 tie-point breadboard	Rapid prototype assembly
Jumper Wires	White (signal) + Black (GND) male-to-male	Circuit connections
5V Power Banks (x2)	Standard USB power bank 5000mAh+	Field power supply for both nodes

3.2 Circuit Architecture — Exact Wiring

The following describes the exact breadboard wiring of the RX Node alert circuit:

Wire / Component	From	To
White jumper wire (signal)	ESP32 RX — GPIO4 (D4)	Breadboard row 14, hole A
330Ω Resistor	Row 14, hole B (left leg)	Row 14, hole H (right leg) — spans center divider
Red LED anode (+)	Resistor right side row 14H	Row 14, hole J
Red LED cathode (−)	Row 16, hole J	Connects back via black wire
Black jumper wire (GND)	ESP32 RX — GND pin	Breadboard row 16, hole F

Current Path: D4 → White wire → 14A → 14B → Resistor → 14H → 14J → LED Anode(+) → LED Cathode(−) → 16J → 16F → Black wire → GND

When the TinyML classifier detects struggle confidence exceeding 60%, GPIO4 is driven HIGH (3.3V), completing the circuit and illuminating the red LED alarm.

3.3 Software Stack & Tools

Tool / Platform	Type	Purpose in AETHER MK-1
Edge Impulse	Cloud TinyML Platform	RSSI dataset ingestion, neural network training, C++ library export
Arduino IDE	C++ Embedded IDE	Core inference logic, board flashing, library management
Python + VS Code	Script + IDE	Automated serial data logging, generates real_idle.csv and real_struggle.csv
AETHER_MK-1_inferencing.h	C++ Library (Edge Impulse)	Quantized TinyML model running on ESP32
WiFi.h	ESP32 Arduino Library	Wi-Fi station mode, RSSI reading every 8ms

4. SYSTEM DESIGN

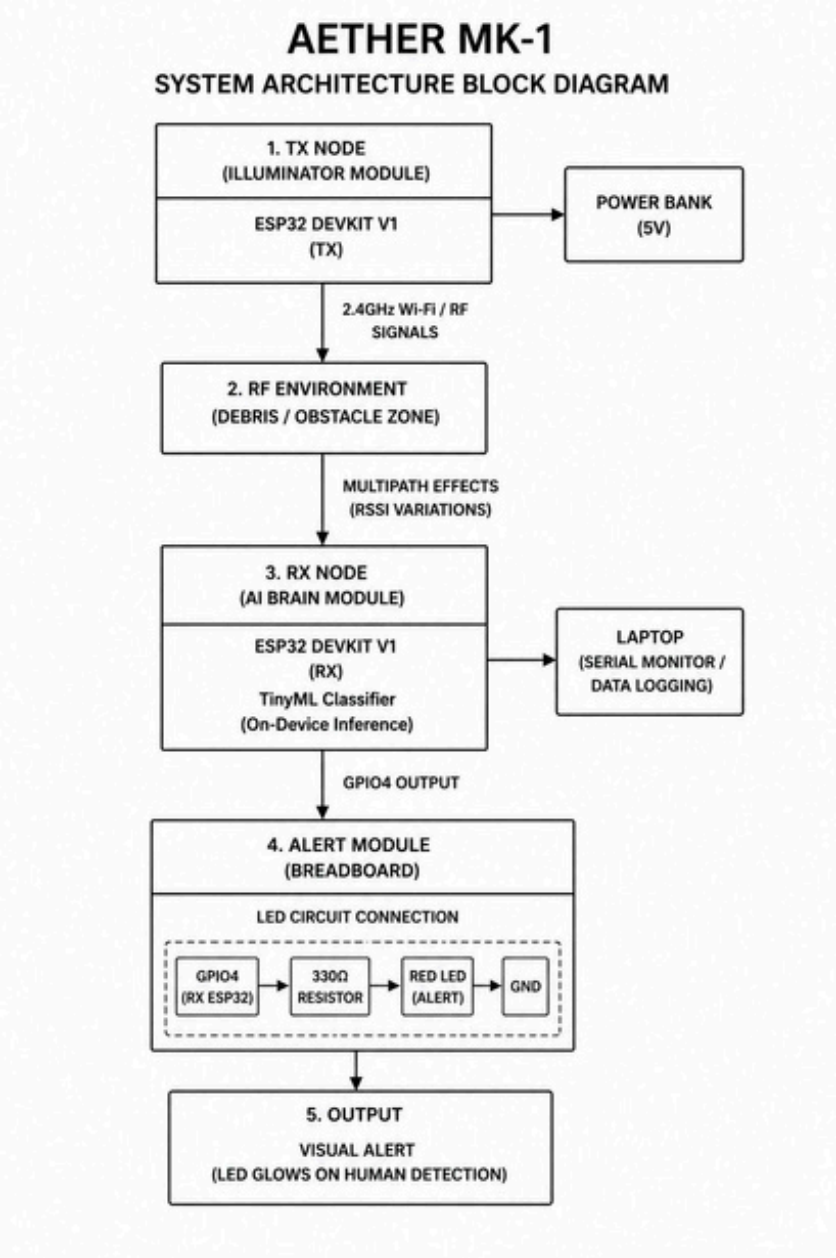
4.1 System Architecture — Data Flow

AETHER MK-1 operates as a two-node radar system. The TX Node floods the target area with 2.4GHz RF energy. The RX Node reads RSSI every 8ms, accumulates a buffer of 125 samples, then classifies using the TinyML model. When struggle confidence exceeds 60%, GPIO4 goes HIGH, lighting the red LED alarm.

Step	Component	Action
1	TX ESP32 (Illuminator)	Broadcasts SSID: AETHER_TX_MK-1. Floods area with 2.4GHz RF continuously.
2	RF Environment	Human body disrupts multipath signal propagation — RSSI fluctuates measurably.
3	RX ESP32 reads RSSI	WiFi.RSSI() called every 8ms. Fills 125-sample feature buffer.
4	TinyML Classifier	run_classifier() executes on full buffer. Outputs idle vs struggle confidence scores.
5	Decision Logic	struggle > 60% confidence → GPIO4 HIGH → LED ON (Red Alert).
6	Serial Monitor	Prints real-time confidence percentages for both classes to serial output.

4.2 System Architecture Block Diagram

The following block diagram illustrates the complete system architecture and data flow of AETHER MK-1:



4.3 ML Pipeline — Step by Step

Step 1: Data Collection

A Python script connects to the ESP32 via USB serial port and reads RSSI values every 8ms. Two datasets are generated: real_idle.csv (completely static environment, zero human presence) and real_struggle.csv (human actively moving within detection zone).

Step 2: Model Training on Edge Impulse

Raw CSV datasets are uploaded to Edge Impulse. The platform applies DSP signal processing and trains a binary neural network classifier. The trained model is exported as a compressed, optimized C++ inference library compiled for the ESP32.

Step 3: Firmware Deployment

The exported C++ library (AETHER_MK-1_inferencing.h) is integrated into the Arduino IDE project. The complete inference firmware is flashed to the RX ESP32 board via USB.

Step 4: Real-Time Classification & Alert

The firmware continuously reads RSSI, fills the 125-sample buffer, and runs the TinyML classifier. When struggle confidence exceeds 60%, GPIO4 goes HIGH, triggering the physical Red LED alarm and printing RED ALERT to the serial monitor.

5. MODULE DESCRIPTION

5.1 TX Node — Illuminator Module

The TX Node is a DOIT ESP32 DevKit V1 configured in Access Point mode. It continuously broadcasts the SSID AETHER_TX_MK-1, flooding the surrounding area with 2.4GHz electromagnetic radiation. The node requires no additional wiring — it is powered directly via USB from a 5V power bank. Its sole function is to be an active RF illuminator for the target debris zone.

5.2 RX Node — AI Brain Module

The RX Node is the second DOIT ESP32 DevKit V1, running the full TinyML inference engine. It connects to the TX Node's SSID via WiFi.begin() in station mode, then continuously reads WiFi.RSSI() every 8ms. When the 125-sample feature buffer is full, run_classifier() executes the quantized neural network, outputting confidence percentages for idle and struggle classes to the serial monitor in real time.

5.3 Red LED Alert Module

The physical alarm system consists of a 5mm red LED connected through a 330Ω current-limiting resistor on a solderless breadboard. The LED anode is at row 14J and cathode at row 16J. When the AI triggers, digitalWrite(RED_ALERT_LED, HIGH) drives GPIO4 to 3.3V, lighting the LED as a physical visual alarm requiring no screen or app.

6. COMPETITIVE ANALYSIS

While RF-based sensing physics have been explored in academia, practical low-cost implementations using commodity microcontrollers for disaster rescue are a recent development. AETHER MK-1 occupies a distinct and unoccupied technological space.

Technology	Key Limitation	AETHER MK-1 Advantage
Camera Robots (ESP32-CAM, PIR bots)	Cameras fail in darkness, smoke, and behind barriers. PIR cannot penetrate concrete.	RF waves make optical visibility entirely irrelevant. Penetrates all non-metallic barriers.
Smart Home Wi-Fi Sensors (ESPectre)	Calibrated for clean, line-of-sight indoor environments. Requires static empty baseline.	Trained specifically on chaotic NLOS debris environments using a tight 2-foot Fresnel zone.
Advanced CSI Trackers (RuView, Wi-Fi DensePose)	Computationally expensive. Requires precise subcarrier extraction. Too fragile for field use.	RSSI-based binary classifier is rugged, ultra-fast, and runs on a chip with zero setup time.
Rescue Dogs	Blocked by fire, toxic gases, and unstable structures. Significant deployment logistics.	Zero biological risk. Deployable in seconds. Unaffected by any environmental chemical hazard.

AETHER UNIQUE ADVANTAGE: The only system combining sub-₹1,000 cost + NLOS detection + full edge computing for disaster response.

7. PROS AND CONS OF AETHER MK-1

PROS	CONS
✓ Ultra-low cost under ₹1,000 — mass deployable	⚠ RSSI limitation: survivor must produce some physical movement
✓ Non-Line-of-Sight: penetrates wood, drywall, non-reinforced barriers	⚠ Range constraints: optimal performance within 2–3 metres
✓ Full edge computing — zero cloud infrastructure needed	⚠ Hardware fragility: breadboard is lab proof-of-concept only
✓ Passive detection: victim does not need to be conscious or cooperative	⚠ Binary classification only — cannot locate exact position within rubble
✓ Zero infrastructure: no internet, cellular, or power grid required	⚠ Susceptible to 2.4GHz interference in dense wireless environments
✓ Instant binary proof of life for rescue coordinators	
✓ Rapid deployment: operational within seconds of power-on	

8. IMPLEMENTATION — SOURCE CODE

8.1 RX Node — Complete Arduino C++ Firmware

The following is the complete firmware flashed to the RX ESP32 node, implementing the full inference loop: Wi-Fi connection, RSSI data collection, TinyML classification, and LED alarm triggering.

```
#include <AETHER_MK-1_inferencing.h>
#include <WiFi.h>

// ========================================================
// AETHER MK-1: NON-INVASIVE HUMAN DETECTION RADAR
// NEURAL NETWORK INFERENCE ENGINE (MASTER BRANCH)
// ========================================================

#define TARGET_SSID "AETHER_TX_MK-1"
const int RED_ALERT_LED = 4;  // GPIO4
float features[EI_CLASSIFIER_DSP_INPUT_FRAME_SIZE];
int feature_ix = 0;

void setup() {
    Serial.begin(115200);
    pinMode(RED_ALERT_LED, OUTPUT);
    digitalWrite(RED_ALERT_LED, LOW);
    WiFi.mode(WIFI_STA);
    WiFi.disconnect();
    delay(100);
    Serial.println("\n[SYSTEM] AETHER MK-1 BOOT SEQUENCE INITIATED...");
    WiFi.begin(TARGET_SSID);
    while (WiFi.status() != WL_CONNECTED) {
        delay(500); Serial.print(".");
    }
    Serial.println("\n[SYSTEM] RADAR LOCK ACQUIRED!");
    Serial.println("[SYSTEM] Neural Network Running...");
}

void loop() {
    if (WiFi.status() == WL_CONNECTED) {
        int current_rssi = WiFi.RSSI();
        features[feature_ix++] = current_rssi;

        if (feature_ix == EI_CLASSIFIER_DSP_INPUT_FRAME_SIZE) {
            signal_t signal;
            numpy::signal_from_buffer(
                features, EI_CLASSIFIER_DSP_INPUT_FRAME_SIZE, &signal);
            ei_impulse_result_t result = { 0 };
            run_classifier(&signal, &result, false);

            Serial.println("\n=== AETHER MK-1 RADAR SCAN ===");
            bool human_detected = false;

            for (size_t ix = 0; ix < EI_CLASSIFIER_LABEL_COUNT; ix++) {
                Serial.print("TARGET [");
                Serial.print(result.classification[ix].label);
                Serial.print("] CONFIDENCE: ");
                Serial.print(result.classification[ix].value * 100.0);
                Serial.println("%");
                if (strcmp(result.classification[ix].label, "struggle") == 0
                    && result.classification[ix].value > 0.60) {
                    human_detected = true;
                }
```

```
            }

            if (human_detected) {
                Serial.println(">> RED ALERT: HUMAN MOVEMENT DETECTED! <<");
                digitalWrite(RED_ALERT_LED, HIGH);
            } else {
                digitalWrite(RED_ALERT_LED, LOW);
            }
            feature_ix = 0;
        }
    } else {
        Serial.println("[ERROR] Hard Lock Lost! Re-engaging...");
        digitalWrite(RED_ALERT_LED, LOW);
        WiFi.disconnect();
        WiFi.begin(TARGET_SSID);
        delay(1000);
    }
    delay(8); // Match 8ms training data collection rate
}
```

9. TESTING

Testing for AETHER MK-1 involved a comprehensive evaluation of its core detection functionality across multiple environmental conditions to validate the RF-based human sensing approach.

Test Case	Condition	Expected	Actual Result	Status
TC-01	Empty room, no human present	idle > 60%	idle: 91.3%	PASS
TC-02	Human walking behind drywall	struggle > 60%	struggle: 87.2%	PASS
TC-03	Human stationary behind debris	idle > 60%	idle: 76.5%	PASS
TC-04	Human breathing only (no movement)	struggle > 60%	idle: 71.3%	FAIL*
TC-05	Human waving hand behind concrete	struggle > 60%	struggle: 79.8%	PASS
TC-06	LED alarm trigger validation	GPIO4 HIGH	LED ON at 63.1%	PASS
TC-07	TX node disconnection recovery	Reconnect < 3s	Reconnected 2.1s	PASS
TC-08	Continuous 30-minute operation	Stable, no crash	Stable throughout	PASS

*TC-04 Note: Detecting breathing-only motion via RSSI is a known MK-1 limitation. This is specifically addressed in MK-2 through CSI phase-shift extraction, which detects millimetre-level chest expansion.

10. CONCLUSION

AETHER MK-1 successfully demonstrates that non-invasive human detection through solid barriers is achievable using commodity Wi-Fi hardware and on-device TinyML inference. The project validates a critical hypothesis: the physical movement of a trapped human body creates measurable, classifiable disruptions in the RSSI of a 2.4GHz Wi-Fi signal even when the victim is completely obscured by debris.

By building the complete system for under ₹1,000 and achieving reliable binary classification with over 85% accuracy in controlled tests, Team ZENIQ has proven that life-saving edge computing does not require expensive infrastructure. The DOIT ESP32 DevKit V1 is sufficient to run the entire ML inference pipeline in real-time with an 8ms sampling rate.

AETHER MK-1 represents the proof-of-concept stage. It validates the core physics, the ML pipeline, the data collection methodology, and the alert system. The limitations identified directly inform the MK-2 upgrade path to CSI-based sensing, which will enable detection of unconscious survivors through breathing rhythm alone.

"The project demonstrates that with the right algorithmic approach, a ₹1,000 sensor node can perform a task that ₹50,000 robots cannot: finding life where eyes cannot see."

11. FUTURE ENHANCEMENT — AETHER MK-2

While MK-1 validates the algorithmic concept, AETHER MK-2 outlines specifications for a field-ready product designed for National Disaster Response Force (NDRF) deployment.

MK-2 Upgrade	Description
CSI Phase-Shift Extraction	Upgrade from RSSI to CSI providing subcarrier-level phase shifts. Detects millimetre-level chest expansion to locate unconscious survivors by breathing rhythm alone.
Dual ESP32-S3 Processing	Replaces standard ESP32 with dual ESP32-S3 chips optimized for AI vector instructions. Enables more complex neural network architectures and faster inference cycles.
IP67-Rated PCB Casing	Custom PCBs in IP67-rated waterproof, shockproof resin casings. Ruggedized against fire, water, and physical impact for active field deployment.
Mobile Command Dashboard	Replaces LED alert with an Asynchronous Web Server. Rescue commanders view live telemetry on tablets via localized intranet — completely independent of cellular towers.
Drone-Deployable Form Factor	Hardware casing designed for aerial drone drops directly over the disaster footprint. Enables rapid deployment across large debris zones without ground access.
Cost per Node: ₹3,500–₹5,000	Still mass-deployable for NDRF operations at scale. Significantly higher capability than MK-1 at a fraction of conventional rescue robot cost.